

TBI-GEU Summer Internship Program 2026

Week 1 — Dataset Identification & Onboarding Confirmation

Intern ID: 26101136 | AI-Enabled Business Intelligence

Environment Setup Confirmation

Platform: Google Colab (colab.research.google.com) | Python 3.10+

Test code executed successfully in a new Colab notebook:

```
import pandas as pd
import matplotlib.pyplot as plt
print('setup ok')

# Output: setup ok
```

Dataset Loaded: Batch Yield Distillation Log (Custom)

Dataset created and loaded into Colab environment. Source: Custom simulated dataset based on realistic Himalayan herbal distillation parameters. File: batch_yield_log.csv

df.shape

```
(624, 10)
```

df.columns

```
Index(['Batch_ID', 'Plant_Variety', 'Harvest_Season', 'Altitude_Zone',
       'Input_Kg', 'Distillation_Duration_hrs', 'Temperature_C',
       'Output_ml', 'Yield_Efficiency_pct', 'Quality_Grade'], dtype='object')
```

df.head() — First 5 rows

Batch_ID	Plant_Variety	Season	Altitude	Input_Kg	Duration_hrs	Temp_C	Output_ml	Yield%	Grade
B001	Rosemary	Summer	High (>2500m)	50	4.5	78	1820	3.64	A
B002	Chamomile	Monsoon	Mid (1500-2500m)	45	5.0	75	1530	3.40	B
B003	Lavender	Spring	High (>2500m)	60	4.0	80	2340	3.90	A
B004	Mint	Summer	Low (<1500m)	40	3.5	82	1280	3.20	B
B005	Eucalyptus	Winter	Mid (1500-2500m)	55	5.5	76	1750	3.18	C

df.info() — Column types and null counts

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 624 entries, 0 to 623
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Batch_ID                             624 non-null   object
1   Plant_Variety                        624 non-null   object
2   Harvest_Season                       624 non-null   object
3   Altitude_Zone                       624 non-null   object
4   Input_Kg                             624 non-null   float64
5   Distillation_Duration_hrs           624 non-null   float64
6   Temperature_C                       624 non-null   float64
```

```
7   Output_ml          624 non-null    float64
8   Yield_Efficiency_pct 624 non-null    float64
9   Quality_Grade       624 non-null    object
dtypes: float64(5), object(5) | memory usage: 48.8 KB
Missing values: 0
```